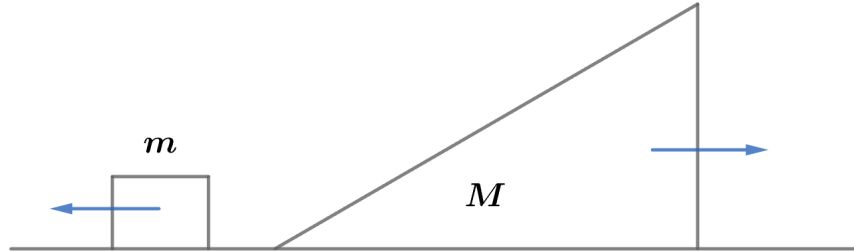


2010 F=ma Exam: Problem 16

Kevin S. Huang



After the block leaves the inclined plane, the system is equivalent to an elastic collision (since energy and momentum are both conserved). The velocity of the block is given by the elastic collision equation,

$$v_1 = \frac{m - M}{m + M} v_0$$

The speed is the magnitude of the velocity,

$$v = |v_1| = \frac{M - m}{m + M} v_0$$

so the answer is E.